

Microcontroller: STM32C031C6

Display module: ST7735 TFT LCD (SPI Interface)

Input controls: 4 x 6mm tactile push buttons

Operating voltage: 3.3V logic & power

Display pin mapping

ST7735 pin	STM32 pin	Description
GND	GND	Ground Reference
VCC	3V3	3.3V Power Supply
SCL	PA5	SPI1 Serial Clock (SPI1_SCK)
SDA	PA7	SPI1 Data Input (SPI1_MOSI)
RES	PB0	Hardware Display Reset (GPIO Out)
DC	PB1	Data / Command Selector (GPIO Out)
CS	PB2	Chip Select (GPIO / SPI1 CS)
BLK	3V3	Backlight Power (3.3V or PWM GPIO)

Tactile switch pin mapping

button	STM32 pin	Configuration mode
UP	PD0	GPIO Input with Internal Pull-Up (active: LOW, idle: HIGH)
DOWN	PD1	-
LEFT	PD2	-
RIGHT	PD3	-

STM32CubeMX notes

- A. SPI1 Peripheral Configuration:
 - a. Mode: Master Transmit Only
 - b. Data Size: 8-bit
 - c. Clock Polarity (CPOL): Low (CPOL = 0)
 - d. Clock Phase (CPHA): 1 Edge (CPHA = 0)
 - e. Baud Rate Prescaler: Configured for ≤ 15 -20 MHz clock speed
- B. GPIO Output Pins:
 - a. Mode: Output Push-Pull
 - b. Pull: No Pull
 - c. Speed: High Speed
- C. GPIO Input Pins (Tactile Switches):
 - a. Mode: Input
 - b. Pull: Internal Pull-Up (GPIO_PULLUP)